2.4 Ordered Rings and the Rules for Inequalities

Section 2.3 made $\mathbb{Z}$ a commutative ring and proved the rules for equations from (F1)–(F9). Inequalities need the same treatment. In school, $-3 < -2$ is read off a number line; here there is no line, only classes of pairs. This section lists four order axioms, shows that an order is the same thing as a set of positive elements, orders $\mathbb{Z}$, and derives the rules for inequalities in every ordered ring. Each rule will come from one observation: $a < b$ is a statement about the single element $b - a$.

2.4.1 Ordered rings

The integers carry an order tied to the operations: adding the same number to both sides keeps an inequality, and a product of positives is positive. We want every rule for inequalities to follow from a short list of axioms, as for equations. The list is the order axioms (O1)–(O4) of *Analysis on the Real Line*, Chapter 1, with "field" relaxed to "commutative ring".

An ordered ring is a ring with a strict order that adding respects, in which a product of positives is positive.

Definition 2.4.1. Ordered Ring, Ordered Field

An **ordered commutative ring** is a commutative ring R (Definition 2.3.1) together with a relation $<$ on R such that, **for all** $a, b, c \in R$:

(O1) (trichotomy) **exactly one** of $a < b$, $a = b$, $b < a$ holds;

(O2) (transitivity) if $a < b$ and $b < c$, then $a < c$;

(O3) (addition) if $a < b$, then $a + c < b + c$;

(O4) (multiplication) if $0 < a$ and $0 < b$, then $0 < ab$.

An **ordered field** is a field that is an ordered commutative ring. We write $b > a$ for $a < b$, and $a \leq b$ for "$a < b$ or $a = b$" (likewise $b \geq a$). An element a is **positive** if $a > 0$, **negative** if $a < 0$, and **non-negative** if $a \geq 0$.

In words: any two elements compare in exactly one way, so $a < a$ never holds; comparisons chain; we may add the same element to both sides; and (O4), the only clause about multiplication, speaks only of **positive** elements.

The degenerate case: the zero ring $\{0\}$ with the **empty** relation is an ordered commutative ring. Only the middle option of (O1) holds, and (O2)–(O4) are vacuous. There $1 = 0$, so any rule concluding $1 > 0$ must assume $1 \neq 0$.

Checking (O1)–(O4) directly means handling pairs of elements. It is easier to say which elements are positive and let $a < b$ mean that $b - a$ is positive. The next proposition says which "positive sets" give an order, and that every order arises so. We call the move **order from a positive set**.

Proposition 2.4.1. Order from a Positive Set

Let R be a commutative ring and $P \subseteq R$ a subset such that:

(P1) if $a, b \in P$, then $a + b \in P$ and $ab \in P$;

(P2) **for every** $a \in R$, **exactly one** of $a \in P$, $a = 0$, $-a \in P$ holds.

Define $a < b$ to mean $b - a \in P$. Then $(R, <)$ is an ordered commutative ring (Definition 2.4.1), and $P = \{a \in R : a > 0\}$.

Conversely, if $(R, <)$ is an ordered commutative ring, then $P := \{a \in R : a > 0\}$ satisfies (P1) and (P2), and for all $a, b \in R$, $a < b$ holds if and only if $b - a \in P$.

The labels (P1), (P2) are local; they are not the clauses of a partition (*Sets and Functions*, Section 5.2) or Peano's axioms (Chapter 1).

Idea. Each order axiom is a statement about one difference. (O1) is (P2) applied to $b - a$, since $-(b-a) = a-b$. (O2) is closure under sums, since $(c-b)+(b-a) = c-a$. (O3) holds since $(b + c) - (a + c) = b - a$. (O4) is closure under products. The algebra is uniqueness of negatives.

Proof. We first record three identities in R, for $a, b, c \in R$:

$$-(b - a) = a - b, \qquad (c - b) + (b - a) = c - a,$$
$$(b + c) - (a + c) = b - a.$$

For the first, $(b - a) + (a - b) = (b + (-b)) + ((-a) + a) = 0$ by (F1), (F2), (F4), and Proposition 2.3.2 (a) gives the identity. The second is regrouping by (F1)–(F4). For the third, $(a + c) + ((-a) + (-c)) = 0$ likewise, so $-(a + c) = (-a) + (-c)$ by Proposition 2.3.2 (a), and the c's cancel. Also $-0 = 0$ (as $0 + 0 = 0$), so $a - 0 = a$; and $b - a = 0$ if and only if $b = a$: one direction is (F4), and $b - a = 0$ gives $b = (b - a) + a = 0 + a = a$.

Now let P satisfy (P1), (P2), and let $a < b$ mean $b - a \in P$.

(O1) Apply (P2) to $b - a$: exactly one of $b - a \in P$, $b - a = 0$, $-(b - a) \in P$ holds. By the facts above, these say $a < b$, $a = b$ and $a - b \in P$, that is, $b < a$. So exactly one of the three holds.

(O2) If $b - a \in P$ and $c - b \in P$, then $c - a = (c - b) + (b - a) \in P$ by (P1).

(O3) If $b - a \in P$, then $(b + c) - (a + c) = b - a \in P$.

(O4) Since $a - 0 = a$, we have $0 < a$ if and only if $a \in P$. So $0 < a$ and $0 < b$ give $a, b \in P$, hence $ab \in P$ by (P1), that is, $0 < ab$. The same equivalence says $P = \{a \in R : a > 0\}$.

Conversely, let $(R, <)$ be an ordered commutative ring and $P := \{a : a > 0\}$. If $a < b$, then $a + (-a) < b + (-a)$ by (O3), that is, $0 < b - a$. If $0 < b - a$, then $0 + a < (b - a) + a = b$ by (O3), that is, $a < b$. This is the last claim. For (P1), let $a, b > 0$. Then $0 + b < a + b$ by (O3), so $0 < b < a + b$ and $a + b > 0$ by (O2); and $ab > 0$ by (O4). For (P2), let $a \in R$. By (O1) exactly one of $0 < a$, $a = 0$, $a < 0$ holds, and by the last claim $a < 0$ holds exactly when $0 - a = -a$ is in P. This proves the proposition.

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So an order on a commutative ring is the same thing as a positive set. For $\mathbb{Z}$, the natural one is the classes $[(a, b)]$ with $b < a$. Section 2.1 warned that "$b < a$" is a statement about **one**

representative; we now check that it is a statement about the class.

2.4.2 The order of the integers

Definition 2.4.2. Order on the Integers

Let $S := \{(a,b) \in \mathbb{N} \times \mathbb{N} : b < a\}$, and

$$P_{\mathbb{Z}} := \{[(a,b)] : (a,b) \in S\} \subseteq \mathbb{Z}.$$

For $x, y \in \mathbb{Z}$, define $x < y$ to mean $y - x \in P_{\mathbb{Z}}$.

Well-definedness. By Theorem 2.2.1 (c), with $X = \mathbb{N} \times \mathbb{N}$, it suffices that $(a,b) \sim (a',b')$ and $b < a$ imply $b' < a'$. Write $a = b + k$ with $k \neq 0$ (Proposition 1.3.1 (a)). Then $a + b' = b + a'$ reads $(b' + k) + b = a' + b$ (Theorem 1.2.3 (b), (c)), and canceling b (Proposition 1.3.1 (b)) gives $a' = b' + k$, so $b' < a'$. Hence, for **every** pair,

$$[(a,b)] \in P_{\mathbb{Z}} \iff b < a.$$

So "positive" may be tested on any representative, as Section 2.1's warning required.

Examples. $[(5,2)] \in P_{\mathbb{Z}}$, since $2 < 5$; this is the integer 3. $[(2,5)] \notin P_{\mathbb{Z}}$ and $[(4,4)] \notin P_{\mathbb{Z}}$, since neither $5 < 2$ nor $4 < 4$. To compare $x = [(3,5)]$ and $y = [(4,1)]$: $y - x = [(4,1)] + [(5,3)] = [(9,4)]$, and $4 < 9$, so $x < y$, as "$-2 < 3$" leads us to expect. The degenerate pair $(0,0)$ is not in S, so $0 \notin P_{\mathbb{Z}}$: zero is not positive.

Quick check. Is $[(7,10)] < [(2,4)]$?

Solution. Compute $[(2,4)] - [(7,10)] = [(2,4)] + [(10,7)] = [(12,11)]$ by Definition 2.2.1. Since $11 < 12$, the difference lies in $P_{\mathbb{Z}}$, so yes: $[(7,10)] < [(2,4)]$, which is $-3 < -2$.

Recall from Section 2.2 that $\mathbb{Z}_{>0} := \{n \in \mathbb{N} : n \neq 0\}$, a set of natural numbers, inside $\mathbb{Z}$ through the identification $n = \iota(n)$. The next theorem shows that it is exactly the set of positive integers.

Theorem 2.4.2. The Integers Form an Ordered Ring

$\mathbb{Z}$, with the operations of Definition 2.2.1 and the relation $<$ of Definition 2.4.2, is an ordered commutative ring, and its positive elements are exactly $P_{\mathbb{Z}}$. Moreover:

(a) for $m, n \in \mathbb{N}$, $\iota(m) < \iota(n)$ if and only if $m < n$;

(b) $\mathbb{N} = \{x \in \mathbb{Z} : x \geq 0\}$, and $\mathbb{Z}_{>0} = \{x \in \mathbb{Z} : x > 0\}$;

(c) (**discreteness**) there is no $k \in \mathbb{Z}$ with $0 < k < 1$; hence, for $m, n \in \mathbb{Z}$ with $m < n$, $m + 1 \leq n$.

Idea. By Proposition 2.4.1 it is enough to check (P1) and (P2) for $P_{\mathbb{Z}}$, each a fact about $\mathbb{N}$, reached by reducing to the level below (Example (Method: reduce to the level below)). For products, write $a = b + k$, $c = d + l$: the product's two coordinates differ by exactly $kl \neq 0$. (P2) is trichotomy in $\mathbb{N}$. Parts (a)–(c) follow from the positive classes being the $[(k,0)]$ with

$k \neq 0$.

Proof. By Theorem 2.3.1, $\mathbb{Z}$ is a commutative ring. We check (P1) and (P2) of Proposition 2.4.1 for $P = P_{\mathbb{Z}}$, using the criterion $[(a,b)] \in P_{\mathbb{Z}} \iff b < a$.

(P1), sums. Let $x = [(a,b)]$ and $y = [(c,d)]$ with $b < a$ and $d < c$. Then $x+y = [(a+c, b+d)]$, and

$$b + d < a + d < a + c$$

by Proposition 1.3.1 (c) twice (with commutativity), so $b + d < a + c$ by transitivity (Theorem 1.3.2 (a)). Hence $x + y \in P_{\mathbb{Z}}$.

(P1), products. With the same x, y, write $a = b + k$ and $c = d + l$ with $k, l \neq 0$ (Proposition 1.3.1 (a)). Then $xy = [(ac + bd, ad + bc)]$, and expanding by Theorem 1.2.3,

$$ac + bd = (b + k)(d + l) + bd = bd + bl + kd + kl + bd,$$
$$ad + bc = (b + k)d + b(d + l) = bd + kd + bd + bl.$$

So $ac + bd = (ad + bc) + kl$. Since $k \neq 0$ and $l \neq 0$, $kl \neq 0$ by Proposition 1.3.1 (d), and therefore $ad + bc < ac + bd$ by Proposition 1.3.1 (a). Hence $xy \in P_{\mathbb{Z}}$.

(P2). Let $x = [(a,b)]$. Then $x \in P_{\mathbb{Z}}$ if and only if $b < a$. Next, $x = [(0,0)]$ if and only if $(a,b) \sim (0,0)$ (Sets and Functions, Theorem 5.2.1 (b)), that is, $a + 0 = b + 0$, that is, $a = b$. Finally, $-x = [(b,a)] \in P_{\mathbb{Z}}$ if and only if $a < b$. By trichotomy in $\mathbb{N}$ (Theorem 1.3.2 (a)), exactly one of $b < a$, $a = b$, $a < b$ holds. So exactly one of $x \in P_{\mathbb{Z}}$, $x = 0$, $-x \in P_{\mathbb{Z}}$ holds.

By Proposition 2.4.1, $(\mathbb{Z}, <)$ is an ordered commutative ring with positive set $P_{\mathbb{Z}}$.

(a) By Definition 2.2.1, $\iota(n) - \iota(m) = [(n,0)] + [(0,m)] = [(n,m)]$, which lies in $P_{\mathbb{Z}}$ if and only if $m < n$.

(b) If $x = [(a,b)] > 0$, then $a = b + k$ with $k \neq 0$ (Proposition 1.3.1 (a)), and $x = [(k,0)] = \iota(k)$ by Proposition 2.1.2 (a). Conversely, for $k \neq 0$, $\iota(k) = [(k,0)]$ is positive since $0 < k$ (Theorem 1.3.2 (d)). So the positive integers are the $\iota(k)$ with $k \in \mathbb{Z}_{>0}$; adding $0 = \iota(0)$ gives the first equation.

(c) If $0 < k$, then $k = \iota(m)$ with $m \neq 0$ by (b), so $1 \leq m$ (Theorem 1.3.2 (d)) and $1 \leq k$ by (a); then $k < 1$ would give $1 < 1$ by (O2), or contradict (O1) directly. Now let $m < n$. Then $n - m > 0$ (Proposition 2.4.1), so $n - m \geq 1$, and adding m by (O3) gives $n \geq 1 + m$. This proves the theorem.

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With the identification of Section 2.2, (a) says that the order of $\mathbb{Z}$ **extends** the order of $\mathbb{N}$.

Two non-examples by minimal change, each breaking exactly one order axiom:

- Replace $<$ by $\leq$. (O2)–(O4) still hold, by the axioms for $<$ and the cases of equality (for (O4), $ab = 0$ when a factor is 0, by Proposition 2.3.2 (b)). But (O1) fails at $a = b = 0$: both "$0 \leq 0$" and "$0 = 0$" hold. The axioms describe a **strict** order.
- Reverse the order: $a \prec b$ means $b < a$. (O1)–(O3) hold, by the same axioms for $<$. But (O4) fails: $0 \prec -1$, since $0 - (-1) = 1 = [(1,0)] \in P_{\mathbb{Z}}$ (Proposition 2.3.2 (a)), and $(-1)(-1) = 1$ (Proposition 2.3.2 (c)), yet $0 \prec 1$ is false, since $0 - 1 = [(0,1)] \notin P_{\mathbb{Z}}$. So (O4) fixes which way round the order goes.

2.4.3 Rules for inequalities

The rules below hold in every ordered commutative ring: in $\mathbb{Z}$ now, and in $\mathbb{Q}$ once Section 3.2 orders it. One of them needs multiples first.

Multiples. Let R be a commutative ring and $a \in R$. By Theorem 1.2.1, with $X = R$, starting value 0 and $g(n, y) = y + a$ (and $n + 1 = \mathrm{S}(n)$, Definition 1.2.1 (c)), there is exactly one function $\mathbb{N} \to R$, written $n \mapsto n \cdot a$, with

$$0 \cdot a = 0, \qquad (n+1) \cdot a = n \cdot a + a \quad \text{for every } n \in \mathbb{N}.$$

So $1 \cdot a = a$ and $2 \cdot a = a + a$. From here to the end of the section, a dot with a natural number on its left means this multiple, and the product of two elements of R is written by juxtaposition, ab. When $R = \mathbb{Z}$, the two readings agree, by Theorem 1.1.2 (b): $0 \cdot a = 0 = 0a$ by Proposition 2.3.2 (b) and (F6), and $n \cdot a = na$ gives $(n+1) \cdot a = na + a = (n+1)a$, since $(n+1)a = a(n+1) = an + a = na + a$ by (F6), then (F9) with (F7), then (F6) (the sum $n + 1$ is the same in $\mathbb{N}$ and $\mathbb{Z}$ by Proposition 2.2.3 (a)).

Proposition 2.4.3. Rules for Inequalities in an Ordered Ring

Let R be an ordered commutative ring, and let $a, b, c, d \in R$.

(a) $a < b$ if and only if $b - a > 0$, if and only if $-b < -a$.

(b) If $a \le b$ and $c \le d$, then $a + c \le b + d$, and the conclusion is strict if one of the hypotheses is strict.

(c) If $a < b$ and $c > 0$, then $ac < bc$.

(d) If $a < b$ and $c < 0$, then $ac > bc$.

(e) If $a \neq 0$, then $a^2 > 0$. If $1 \neq 0$, then $1 > 0$.

(f) If $ab = 0$, then $a = 0$ or $b = 0$.

(g) If $a \ge 0$ and $b \ge 0$, then $a < b$ if and only if $a^2 < b^2$.

(h) For every $m, k \in \mathbb{N}$, $(m + k) \cdot a = m \cdot a + k \cdot a$. If $1 \neq 0$, then $n \cdot 1 > 0$ for every $n \in \mathbb{N}$ with $n \ge 1$, and $m \cdot 1 < n \cdot 1$ whenever $m, n \in \mathbb{N}$ and $m < n$.

(i) $\le$ is a total order on R (Sets and Functions, Definition 5.3.1), and its strict part is $<$.

Idea (Scratch work). Every part is a comparison, and by Proposition 2.4.1 a comparison $a < b$ is the statement $b - a > 0$. Positive elements are closed under $+$ and $\cdot$. So the plan is always: write the goal as one difference, and build it from given positive elements. For (c):

Givens	Goal
$a < b$, $c > 0$	$ac < bc$
$b - a > 0$, $c > 0$ (Proposition 2.4.1)	$bc - ac > 0$
$(b - a)c > 0$ by (O4)	$bc - ac = (b - a)c$?
(F9), (F6), Proposition 2.3.2 (c)	yes: the gap closes

Part (d) is (c) with $-c$ in place of c; (e) and (f) apply (O4) to whichever of a, $-a$ is positive. We prove (i) first, since (b) and (g) use it.

Proof.

(i) Reflexivity is $a = a$. Antisymmetry: $a \le b$, $b \le a$ and $a \ne b$ give $a < b < a$, so $a < a$ by (O2), against (O1). Transitivity: if $a \le b \le c$, either an equality holds and we substitute, or $a < c$ by (O2); and $a < c$ whenever one step is strict. Totality is (O1). Strict part: $a < b$ implies $a \ne b$ by (O1), and $a \le b$, $a \ne b$ give $a < b$.

(a) The first equivalence is the last claim of Proposition 2.4.1. For the second, $(-a)-(-b) = (-a) + b = b - a$ by Proposition 2.3.2 (a) and (F1), so $-b < -a$ also says $b - a > 0$.

(b) By (O3) (or equality) and (F1), $a + c \le b + c \le b + d$; conclude by (i).

(c) By (a), $b - a > 0$. By (O4), $(b - a)c > 0$. By (F9), (F6) and Proposition 2.3.2 (c), $(b - a)c = bc + (-a)c = bc - ac$. So $bc - ac > 0$, and $ac < bc$ by (a).

(d) By (a), $c < 0$ gives $0 - c = -c > 0$. By (c) applied with $-c$, $a(-c) < b(-c)$, that is, $-(ac) < -(bc)$ by Proposition 2.3.2 (c), and then $bc < ac$ by (a) together with $-(-x) = x$.

(e) If $a > 0$, then $a^2 = aa > 0$ by (O4). If $a < 0$, then $-a > 0$ as in (d), and $a^2 = (-a)(-a) > 0$ by Proposition 2.3.2 (c) and (O4). By (O1) one of the two cases occurs when $a \ne 0$. If $1 \ne 0$, then $1 = 1^2 > 0$.

(f) Contrapositive: let $a, b \ne 0$. By (O1) and (a), some $a' \in \{a, -a\}$ and $b' \in \{b, -b\}$ are positive, so $a'b' > 0$ by (O4). By Proposition 2.3.2 (c), $a'b'$ is ab or $-(ab)$; if $ab = 0$, both are 0, against (O1). So $ab \ne 0$.

(g) Let $a, b \ge 0$. ($\Rightarrow$) If $a < b$, then $b > 0$ by (i), and (c) gives $ab < b^2$. Also $a^2 \le ab$: by (c) if $a > 0$, and by Proposition 2.3.2 (b) if $a = 0$. So $a^2 < b^2$ by (i). ($\Leftarrow$) If $a^2 < b^2$ but not $a < b$, then $b = a$ or $b < a$ by (O1), giving $a^2 = b^2$ or, by ($\Rightarrow$), $b^2 < a^2$; both contradict (O1).

(h) Fix $m \in \mathbb{N}$. We prove the sum law by Theorem 1.1.2 (b) on k, for $P(k)$: "$(m+k)\cdot a = m \cdot a + k \cdot a$". The base case is $(m + 0) \cdot a = m \cdot a + 0 \cdot a$, by Definition 1.2.1 (a), (F3) and $0 \cdot a = 0$. For the step, $P(k)$ gives $(m + (k + 1)) \cdot a = (m + k) \cdot a + a = (m \cdot a + k \cdot a) + a = m \cdot a + (k + 1) \cdot a$, by associativity in $\mathbb{N}$ (Theorem 1.2.3 (c)) with the definition of multiples, $P(k)$, and (F2) with the definition of multiples again.

Now let $1 \ne 0$. By Theorem 1.1.2 (b) on n, for $Q(n)$: "$(n + 1) \cdot 1 > 0$": the base is $1 \cdot 1 = 0 + 1 = 1 > 0$ by (e), and the step is $(n + 2) \cdot 1 = (n + 1) \cdot 1 + 1 > 0$ by (b) and (e). Every $n \ge 1$ in $\mathbb{N}$ is $j + 1$ for some j (Proposition 1.3.1 (a)), so $n \cdot 1 > 0$. Finally, if $m < n$, then $n = m + k$ with $k \ne 0$ (Proposition 1.3.1 (a)), so $k \ge 1$ (Theorem 1.3.2 (d)), and by the sum law and (b),

$$n \cdot 1 = m \cdot 1 + k \cdot 1 > m \cdot 1 + 0 = m \cdot 1.$$

This proves the proposition.

■

The school rules for inequalities between integers are now theorems about $\mathbb{Z}$. The hypothesis $1 \ne 0$ in (e) and (h) cannot be dropped: in the zero ring with the empty order, neither $1 > 0$ nor $1 \cdot 1 > 0$ holds.

Example. Method: turn the comparison into one positive element

Template. To prove an inequality in an ordered ring: (1) rewrite the goal $u < v$ as $v - u > 0$ (Proposition 2.4.1); (2) express $v - u$ by the ring laws as a sum or product of elements known

to be positive (or non-negative); (3) conclude by closure, (P1), or by Proposition 2.4.3 (b), (c); (4) translate back.

When to use it. Any rule for inequalities; comparisons in $\mathbb{Z}$ and, later, in every ordered field.

The usual failure. Multiplying by an element whose sign is unknown; treating $\leq$ as $<$ in (O4); using a picture of the number line as a reason.

Warning. (O4) and (c) are about multiplying by something **positive**. The tempting rule "if $a < b$ and $c < d$, then $ac < bd$" is false in $\mathbb{Z}$: $-3 < -1$ and $-3 < -1$, yet $(-3)(-3) = 9$ and $(-1)(-1) = 1$, and $9 > 1$.

With both smaller sides non-negative, the method gives a correct rule.

Example. Multiplying two inequalities

Let R be an ordered commutative ring, and let $a, b, c, d \in R$ with $0 \leq a < b$ and $0 \leq c < d$. Prove that $ac < bd$.

Solution. By the method, the goal is $bd - ac > 0$. Write the difference through elements of known sign:

$$bd - ac = b(d - c) + (b - a)c.$$

Indeed, $b(d - c) = bd - bc$ and $(b - a)c = bc - ac$, by (F9), (F6) and Proposition 2.3.2 (c); adding, the terms $-bc$ and bc cancel by (F1), (F2), (F4) and (F3), leaving $bd - ac$.

Now the signs. From $0 \leq a$ and $a < b$ we get $0 < b$ (Proposition 2.4.3 (i)), and $d - c > 0$ since $c < d$ (Proposition 2.4.3 (a)). So $b(d - c) > 0$ by (O4). Likewise $b - a > 0$. If $c > 0$, then $(b - a)c > 0$ by (O4); if $c = 0$, then $(b - a)c = 0$ by Proposition 2.3.2 (b). Either way $(b - a)c \geq 0$.

A positive element plus a non-negative one is positive: by Proposition 2.4.3 (b), with the strict hypothesis $0 < b(d-c)$, we get $0 = 0+0 < b(d-c)+(b-a)c = bd-ac$, where $0+0 = 0$ is (F3). By Proposition 2.4.3 (a), $ac < bd$.

The hypotheses $a \geq 0$ and $c \geq 0$ are exactly what make the two summands non-negative; the warning shows what goes wrong without them.

Quick check. Which positive element shows that $a < b$ gives $a + a < b + b$?

Solution. The element $b - a$, taken twice: $(b + b) - (a + a) = (b - a) + (b - a)$ by the ring laws (regrouping by (F1), (F2), with $-(a + a) = (-a) + (-a)$ as in the proof of Proposition 2.4.1). It is a sum of two positive elements, so it is positive by (P1) for the positive set (Proposition 2.4.1, converse), and $a + a < b + b$ by Proposition 2.4.3 (a).

Quick check. Is $\mathbb{Z}$ a field?

Solution. No: $1+1$ is non-zero (it is $> 1 > 0$, where $1 > 0$ by Proposition 2.4.3 (e), as $1 \neq 0$ in $\mathbb{Z}$ by Theorem 2.3.1) and has no inverse. Suppose $(1+1)x = 1$. Since $(1+1)x = x(1+1) = x + x$ by (F6), (F9) and (F7), this says $x + x = 1$. If $x \leq 0$, then $x + x \leq 0 < 1$ by Proposition 2.4.3 (b), (e). If $x > 0$, then $x \geq 1$ by discreteness (Theorem 2.4.2 (c)), so $x + x \geq 1 + 1 > 1$ by Proposition 2.4.3 (b), (e). Both contradict $x + x = 1$ by (O1).

2.4.4 Least elements, and a ring that cannot be ordered

The natural numbers are well ordered (Theorem 1.3.3); the integers are not, since $x - 1 < x$ for every x. But a set of integers bounded below can be shifted into $\mathbb{N}$.

Proposition 2.4.4. Least and Greatest Elements of Sets of Integers

Let $S \subseteq \mathbb{Z}$ be **non-empty**. If S has a lower bound $b \in \mathbb{Z}$ (that is, $b \leq s$ for every $s \in S$), then S has a least element. If S has an upper bound $c \in \mathbb{Z}$, then S has a greatest element.

Idea. This is Sets and Functions, Corollary 3.1.5 on the constructed $\mathbb{Z}$, by the same shift into $\mathbb{N}$, with Theorem 1.3.3 in place of Sets and Functions, Theorem 3.1.3.

Proof. Let b be a lower bound and $T := \{s - b : s \in S\}$, a non-empty set. For $s \in S$, $s \geq b$ gives $s - b \geq 0$ by (O3), so $T \subseteq \{x \in \mathbb{Z} : x \geq 0\} = \mathbb{N}$ by Theorem 2.4.2 (b). By Theorem 1.3.3, T has an element m with $m \leq t$ for every $t \in T$, in the order of $\mathbb{N}$ and so in that of $\mathbb{Z}$ (Theorem 2.4.2 (a)). Then $m = s_0 - b$ for some $s_0 \in S$, so $m + b = s_0 \in S$; and for $s \in S$, $s - b \geq m$ gives $s \geq m + b$ by (O3). So $m + b$ is least in S.

If c is an upper bound, then $-S := \{-s : s \in S\}$ has the lower bound $-c$ (Proposition 2.4.3 (a)), so by the first part it has a least element, which is $-s_1$ for some $s_1 \in S$, by the definition of $-S$. For $s \in S$, $-s_1 \leq -s$ gives $s \leq s_1$, by Proposition 2.4.3 (a), or by $-(-s) = s$ (Proposition 2.3.2 (a)) when $-s_1 = -s$. So s_1 is greatest in S. This proves the proposition. ∎

Both hypotheses are needed: $\mathbb{Z}$ has no least element, and neither has $\emptyset$, which every integer bounds. The rules of Proposition 2.4.3 also show that some rings carry no order at all.

Example. A field with two elements that cannot be ordered

Let $R := \{0, 1\}$, two distinct elements, with addition and multiplication given by the tables

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

Check that R satisfies (F1)–(F9), so that it is a field, and show that no relation $<$ on R makes it an ordered commutative ring.

Solution. The axioms. (F1), (F6): both tables are symmetric. (F3), (F7): the row of 0 in the first table and of 1 in the second repeat the headings. (F4): $-0 = 0$ and $-1 = 1$, since $1 + 1 = 0$. (F8): 1 has inverse 1, since the second table gives the product of 1 and 1 as 1. And $1 \neq 0$.

The three laws with three variables can be checked over the eight triples (a, b, c), or by cases. From the tables, $0x = 0$ and $1x = x$, and $x + y = 1$ exactly when exactly one of x, y is 1. (F2): both sides are 1 exactly when an odd number of a, b, c (one or three) are 1. (F5): both sides are 1 exactly when $a = b = c = 1$. (F9): if $a = 0$, both sides are 0; if $a = 1$, both sides are $b + c$. So R is a field.

No order. If $<$ made R an ordered commutative ring, then, as $1 \neq 0$, Proposition 2.4.3 (h) would give $2 \cdot 1 > 0$. But $2 \cdot 1 = 1 \cdot 1 + 1 = (0 + 1) + 1 = 0$ by the definition of multiples, and $0 > 0$ is forbidden by (O1).

The argument uses only $1 \neq 0$ and $n \cdot 1 = 0$ for some $n \geq 1$. So **no commutative ring with $1 \neq 0$ in which $n \cdot 1 = 0$ for some $n \in \mathbb{N}$, $n \geq 1$, can be ordered**. The integers modulo $n \geq 2$ of *Sets and Functions*, Section 5.2 are such rings (their ring laws taken for granted, as in the warning after Definition 2.3.1), since $n \cdot [1] = [n] = [0]$; *Algebraic Preliminaries*, Chapter 1 rebuilds them on the constructed $\mathbb{Z}$.

2.4.5 Exercises

2.4.5.1 A. Check your understanding

Exercise. A1

Let R be an ordered commutative ring.

(a) Here is a proof that if $a < 0 < b$, then $ab < a^2 + b^2$. Name the axiom, rule or result that justifies each of the five steps. (1) $ab < 0b$. (2) $0b = 0$. (3) $a^2 > 0$ and $b^2 > 0$. (4) $a^2 + b^2 > 0 + 0 = 0$. (5) $ab < a^2 + b^2$.

(b) True or false: "R has no zero divisors." Give a reason.

(c) True or false: "$1 > 0$ in every ordered commutative ring." Give a reason.

(d) Use the method of Example (Method: turn the comparison into one positive element) to show: if $a < b$ and $c \leq 0$, then $ac \geq bc$.

Solution.

(a) (a) Proposition 2.4.3 (c), with $a < 0$ and $b > 0$. (2) Proposition 2.3.2 (b) and (F6).

(3) Proposition 2.4.3 (e), as $a, b \neq 0$ by (O1). (4) Proposition 2.4.3 (b), strict, and (F3). (5) (O2).

(b) True, by Proposition 2.4.3 (f); a general commutative ring may have zero divisors (the warning after Definition 2.3.1).

(c) False: in the zero ring with the empty relation, $1 = 0$. It is true when $1 \neq 0$ (Proposition 2.4.3 (e)).

(d) By the method, compute the difference $ac - bc$. By (F9), (F6) and Proposition 2.3.2 (c), $ac - bc = (a - b)c$; and $a - b = -(b - a)$ (the first identity in the proof of Proposition 2.4.1), so $(a - b)c = (-(b - a))c = (b - a)(-c)$ by Proposition 2.3.2 (c). Here $b - a > 0$ by Proposition 2.4.3 (a). If $c < 0$, then $-c > 0$ (Proposition 2.4.3 (a), as $0 - c = -c$), and the product is > 0 by (O4). If $c = 0$, then $-c = 0$ and the product is 0 by Proposition 2.3.2 (b). So $ac - bc \geq 0$: either $bc < ac$ by Proposition 2.4.3 (a), or $ac - bc = 0$, that is, $ac = bc$. In both cases $bc \leq ac$, that is, $ac \geq bc$.

2.4.5.2 B. Practice

Exercise. B1: The wrong positive set

Let $P' := \{[(a, b)] : a < b\} \subseteq \mathbb{Z}$. Show that membership in P' does not depend on the representative, that P' satisfies (P2) of Proposition 2.4.1, and that it does not satisfy (P1). Which part of (P1) fails?

Solution. Independence of representatives. If $(a, b) \sim (a', b')$ and $b = a + k$ with $k \neq 0$, then $b' + a = a + b' = b + a' = (a + k) + a' = (a' + k) + a$ (Theorem 1.2.3 (b), (c)), so $b' = a' + k$ by cancellation (Proposition 1.3.1 (b)), and $a' < b'$ by Proposition 1.3.1 (a). By Theorem 2.2.1 (c), $[(a, b)] \in P'$ if and only if $a < b$.

(P2). For $x = [(a, b)]$: $x \in P'$ iff $a < b$; $x = 0$ iff $a = b$ (as in the proof of Theorem 2.4.2); $-x \in P'$ iff $b < a$. By trichotomy in $\mathbb{N}$ (Theorem 1.3.2 (a)), exactly one holds.

(P1) fails for products. $[(0, 1)] \in P'$, but, by Definition 2.2.1 with $a = c = 0$ and $b = d = 1$ in $[(ac + bd,\ ad + bc)]$,

$$[(0, 1)][(0, 1)] = [(1, 0)],$$

and $[(1, 0)] \notin P'$, since $1 < 0$ is false. The sum half of (P1) holds, as in the proof of Theorem 2.4.2. So P', the negative integers, fails only closure under products; it would define the reversed order, which breaks exactly (O4).

Exercise. B2: Prove or disprove

Prove or disprove.

(a) In every ordered commutative ring, $a < b$ implies $a^2 < b^2$.

(b) In every ordered commutative ring with $1 \neq 0$, $a^2 + 1 > 0$.

(c) In $\mathbb{Z}$, if $x > 0$ and $xy = x$, then $y = 1$.

(d) In every ordered commutative ring, $a < b$ and $c < d$ imply $a - d < b - c$.

(e) In $\mathbb{Z}$, $x < y$ implies $x + 1 < y$.

Solution.

(a) **False.** In $\mathbb{Z}$, $-3 < 1$, but $(-3)^2 = 9$ and $1^2 = 1$, and $9 > 1$. The true version needs $a \geq 0$ (Proposition 2.4.3 (g)).

(b) **True.** First, $a^2 \geq 0$: if $a = 0$, then $a^2 = 0 \cdot 0 = 0$ by Proposition 2.3.2 (b); otherwise $a^2 > 0$ by Proposition 2.4.3 (e). Next, $1 > 0$ by Proposition 2.4.3 (e), since $1 \neq 0$. By Proposition 2.4.3 (b), with the strict hypothesis $0 < 1$, $0 = 0 + 0 < a^2 + 1$, using (F3).

(c) **True.** By (F9), (F7) and Proposition 2.3.2 (c), $x(y-1) = xy + x(-1) = xy - x$, and $xy - x = x - x = 0$ by the hypothesis and (F4). Since $x > 0$, $x \neq 0$ by (O1). So $y - 1 = 0$ by Proposition 2.4.3 (f), and $y = (y-1) + 1 = 1$ by (F1)–(F4).

(d) **True.** From $c < d$, Proposition 2.4.3 (a) gives $-d < -c$. Adding this to $a < b$ by Proposition 2.4.3 (b), strict, gives $a + (-d) < b + (-c)$, that is, $a - d < b - c$.

(e) **False.** Take $x = 0$ and $y = 1$: then $x < y$ (Proposition 2.4.3 (e), as $1 \neq 0$ in $\mathbb{Z}$ by Theorem 2.3.1), but $x + 1 = 1$, and $1 < 1$ is ruled out by (O1). Discreteness gives only $x + 1 \leq y$ (Theorem 2.4.2 (c)).

2.4.5.3 C. Going deeper

Exercise. C1: Ordered rings with $1 \neq 0$ are infinite

(a) Prove the sum law of Proposition 2.4.3 (h) in full: for every ordered commutative ring R, every $a \in R$ and all $m, k \in \mathbb{N}$, $(m + k) \cdot a = m \cdot a + k \cdot a$. State the property you induct on and justify every equality.

(b) Deduce that every ordered commutative ring with $1 \neq 0$ is infinite (Sets and Functions, Definition 4.1.3).

(c) The set $\{0, 1, 2\}$ with addition and multiplication "modulo 3", given by the tables below, is a commutative ring with $1 \neq 0$ (you may take this for granted). Show that it cannot be ordered.

$+$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\cdot$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

Solution (Hint 1). For (b): what does (h) say about $k \mapsto k \cdot 1$?

Solution (Hint 2). Suppose R has m elements; consider $k \mapsto k \cdot 1$ on $\{1, \ldots, m+1\}$.

Solution.

(a) Fix R, $a \in R$, $m \in \mathbb{N}$, and use Theorem 1.1.2 (b) on k, for $P(k)$: "$(m+k)\cdot a = m\cdot a + k\cdot a$". *Base:* $(m + 0) \cdot a = m \cdot a = m \cdot a + 0 = m \cdot a + 0 \cdot a$, by Definition 1.2.1 (a), (F3) and

$0 \cdot a = 0$. *Step:* assume $P(k)$. Then

$$\begin{aligned}(m + (k+1)) \cdot a &= ((m+k)+1) \cdot a = (m+k) \cdot a + a \\ &= (m \cdot a + k \cdot a) + a = m \cdot a + (k \cdot a + a) \\ &= m \cdot a + (k+1) \cdot a,\end{aligned}$$

by associativity in $\mathbb{N}$ (Theorem 1.2.3 (c)), the recursion for multiples, $P(k)$, (F2), and the recursion again. So $P(k+1)$ holds, and $P(k)$ holds for every k.

(b) Let $1 \neq 0$ in R, and suppose R has m elements for some $m \in \mathbb{N}$ (Sets and Functions, Definition 4.1.2). Let $f\colon \{1, \dots, m+1\} \to R$, $f(k) \coloneqq k \cdot 1$. If $k < l$, then $k \cdot 1 < l \cdot 1$ by Proposition 2.4.3 (h), so $k \cdot 1 \neq l \cdot 1$ by (O1): f is injective. But $\{1, \dots, m+1\}$ has $m + 1 > m$ elements (by its identity map), so Sets and Functions, Corollary 4.1.3 (a) forbids an injection into R. This contradiction shows that R is infinite (Sets and Functions, Definition 4.1.3).

(c) By the definition of multiples and the tables, $1 \cdot 1 = 0 + 1 = 1$, $2 \cdot 1 = 1 + 1 = 2$ and $3 \cdot 1 = 2 + 1 = 0$. An order would give $3 \cdot 1 > 0$ by Proposition 2.4.3 (h), since $1 \neq 0$, that is, $0 > 0$, against (O1). (Alternatively, (b) applies to this finite ring.)

Marking scheme. (10 points) - 1: (a) states the induction property with m and a fixed and k the induction variable. - 2: (a) base case, citing $m + 0 = m$, $0 \cdot a = 0$ and (F3). - 2: (a) inductive step, with associativity in $\mathbb{N}$, the recursion, the hypothesis and (F2) each named. - 2: (b) shows that $k \mapsto k \cdot 1$ is injective, citing (h) and (O1). - 2: (b) derives the contradiction from Sets and Functions, Corollary 4.1.3 (a) with $m + 1 > m$ elements (or any correct count). - 1: (c) computes $3 \cdot 1 = 0$ and concludes. Fatal errors (0 for the step): inducting on m and k at once without fixing one; using $1 > 0$ without $1 \neq 0$; in (b), "the multiples are all different, so there are infinitely many" with no finiteness result cited (Sets and Functions, Corollary 4.1.3, Sets and Functions, Proposition 4.1.6 with $\mathbb{N}$ infinite, or equivalent). A route for (b) composing the inverse of a bijection $\{1, \dots, m\} \to R$ with f earns the same 4 points. So does a route that shows $k \mapsto k \cdot 1$ is an injection $\mathbb{N} \to R$ (2 points, as above), so that its image is an infinite subset of R, and concludes with Sets and Functions, Proposition 4.1.6 (2 points).